

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 12B.1

LATTICE ENERGY AND BORN-HABER CYCLES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	12: Entropy and Energetics
SUB-TOPIC:	12B.1: Lattice Energy and Born-Haber Cycles
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 12B.1

Q1. Define the term lattice energy.

[WCH14/01/Oct22/Q2(a)]

[2 Marks] WCH14/01

Q2. Construct a Born-Haber cycle for sodium chloride and use it to calculate the lattice energy of NaCl. Given:

$\Delta_f H = -411$, $\Delta_{atm} H (Na) = +107$, 1st IE (Na) = +496, $\Delta_{atm} H (Cl) = +122$, 1st EA (Cl) = -349 (all in kJ mol⁻¹). [WCH14/01/Jan22/Q3(b)]

[3 Marks] WCH14/01

SECTION B: WORKED MARK SCHEME — 12B.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Define the term lattice energy. [WCH14/01/Oct22/Q2(a)] The enthalpy change when one mole of a solid ionic compound is formed (1) from its gaseous ions (1) under standard conditions.	[2]
Q2	Q2: Construct a Born-Haber cycle for sodium chloride and use it to calculate the lattice energy of NaCl. Given: $\Delta_f H = -411$, $\Delta_{atm} H (Na) = +107$, 1st IE (Na) = +496, $\Delta_{atm} H (Cl) = +122$, 1st EA (Cl) = -349 (all in kJ mol^{-1}). [WCH14/01/Jan22/Q3(b)] $\Delta_f H = \Delta_{atm} H (Na) + \text{IE}(Na) + \Delta_{atm} H (Cl) + \text{EA}(Cl) + \Delta_{LE} H (1) - 411$ $= 107 + 496 + 122 + (-349) + \Delta_{LE} H (1)$ $\Delta_{LE} H = -411 - 107 - 496 - 122 + 349 = -787 \text{ kJ mol}^{-1} (1)$	[3]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 12B.1**FAQ #1: Why do students lose marks in defining lattice energy?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why do students lose marks in defining lattice energy?

MODEL ANSWER / EXAMINER EXPECTATION:

They often forget to specify that the ions must be in the gaseous state, or they define it for the formation of one mole of gaseous ions rather than one mole of the solid lattice.

FAQ #2: What is the common error when calculating lattice energy from a Born-Haber cycle?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is the common error when calculating lattice energy from a Born-Haber cycle?

MODEL ANSWER / EXAMINER EXPECTATION:

Sign errors are rampant. Ensure that all endothermic processes (atomisation, ionisation energy) are positive and exothermic processes (electron affinity for 1st EA) are negative before solving the equation.